

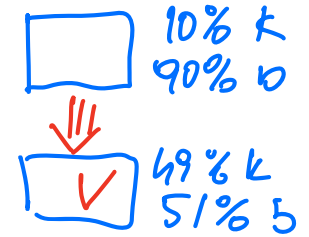
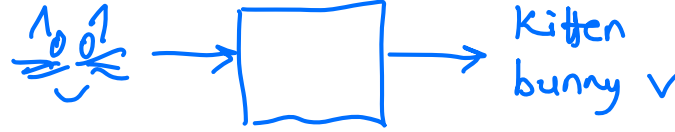
Independence in probabilities, Bayes' nets

$$P(x|y) = \frac{P(x, y)}{P(y)}$$

Models

- A **model** is a way to describe how the world works (or at least, the portion of interest to us)
- You have learned about many models in physics, chemistry, biology, economics
 - The so-called "laws" are very often better described as "models"
- Models are not reality!
 - Even in the best case, they only model the aspects of reality we are interested in
 - We ignore some variables because (a) we are not interested in them or (b) we cannot measure them or (c) we cannot handle the computation...

Probabilistic models



- Models that describe the world through random variables and their relationships
- Sources of randomness:
 - **Intrinsic randomness** in the world we model: eg. our returns at the casino ✓
 - **Lack of knowledge** in predictions: eg. predicting tracks of hurricanes. ✓
 - **Populations** we look at measurements averaged over populations (or repeated samples). Example: predicting ✓
 - **Noisy observations** the world is deterministic, but our observations are affected by random noise ✓
 - **Preference for probabilistic models** sometimes we prefer a probabilistic model because the math works out better. Example: diffusion models for image generation. ✓

How do we use probabilistic models

- The world is described by a set of random variables
- We have some information about the distributions of these variables and the relations between them
 - Some individual probabilities, some joints, some conditionals
- Some variables are known (**evidence**)
- Using the relations between the variables, we answer questions about the world such as:
 - **explanation** - diagnostic reasoning
 - **prediction** - what is the likely value of a variable?
 - **value of information** - what other information do we need to better answer a question?

The best probabilistic model, and why we cannot have it

- Let us say that we describe the world with random variables $X_1, X_2, \dots, X_n$
- If we have the joint probability table $P(X_1, X_2, \dots, X_n)$ all our problems are solved!
 - We can answer any question, for any evidence by simply applying the reasoning process described in the last class: select rows compatible with evidence + marginalize!
- The problem:
 - The joint table is huge! If $n = 10$ and each random variable has 10 possible values, we need 10^{10} probabilities.
 - And this is nothing compared to, eg. the stock market
 - How are we going to build a model this big? Who is going to give us the probabilities?

Two fundamentally different ideas to build a probabilistic model

- Idea 1: Introduce structure
- Idea 2: Use learning

Structured probabilistic models

- We assume that the joint probability table does not contain all independent values. For instance, some of the values can be described by mathematical operations on a smaller number of probabilities
- This can happen, for instance, if some of the variables are **independent** of each other.
- These relationships can be described through some kind of **structure**, often described as a graph
- Examples:
 - Bayesian Networks (directed graphs)
 - Markov Random Fields (undirected graphs)
 - Factor Graphs
 - Probabilistic Circuits, etc.

Learned probabilistic models

- We assume that the probabilities are described by a certain parameterized function

$$P(x_1, x_2, \dots, x_n) = f(x_1, x_2, \dots, x_n; \Theta)$$

Θ^*

- We use data to learn the parameter Θ

- Examples:

- Classical parametric models: eg Gaussian Mixture Models
 - Latent variable models
 - Deep generative models: use deep neural networks for f
 - Variational autoencoders, Normalizing flows, Diffusion models
 - Energy based models
 - Define probability via an unnormalized energy function (we don't need to calculate the normalizing factor Z)

What we will do today?

- One of the simplest ways to simplify probabilistic models:
 - Introduce the concept of independence and conditional independence
 - Create a directed graph (the bayesian network) what describes the set of dependencies between the variables of the model.

Independence

- Two ^{random} variables are independent if:

$$\forall x, y : P(x, y) = P(x)P(y)$$

A handwritten blue diagram above the equation $P(x, y) = P(x)P(y)$. It shows the expression $P(x) \cdot P(y|x)$ with a curved arrow pointing from $P(x)$ to the first P in the equation and another curved arrow pointing from $P(y|x)$ to the $P(y)$ in the equation. There are also two vertical double-headed arrows, one between $P(x)$ and $P(y|x)$, and another between $P(x)$ and $P(y)$.

- This says that their joint distribution factors into a product of two simplest distributions.
- Why is this "independence"? Another way to write this (direct consequence of the definition of a conditional)

$$\forall x, y : \underline{P(x|y)} = \underline{P(x)}$$

We write

$$X \perp\!\!\!\perp Y$$

Independence is a simplifying assumption

- If we have 10 variables with 10 possible values
 - If they are **not independent** we need 10^{10} probability values to describe the joint
 - If they are **independent** we need 100 probability values
- Are there many independent variables in the world?
 - Sure: throwing a dice in Las Vegas and an airport delay in Orlando are usually independent!
 - N fair, independent coin flips are independent
- But any set of variables chosen for a given problem, they are rarely fully independent!

Conditional independence

- X is conditionally independent of Y given Z

$$X \perp\!\!\!\perp Y | Z$$

if and only if

$$\forall x, y, z : P(x, y | z) = P(x | z)P(y | z)$$

or, equivalently

$$\forall x, y, z : P(x | y, z) = P(x | z)$$

Importance of conditional independence

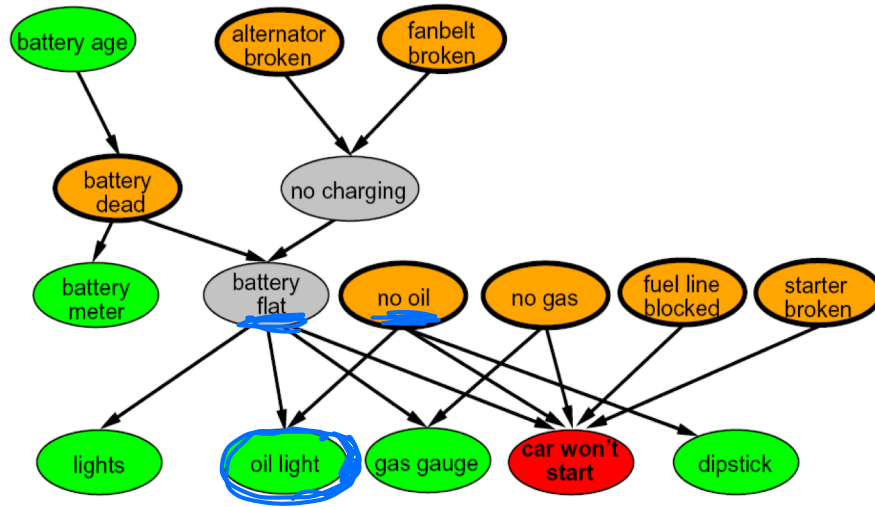
- It is much more likely that we find conditional independence in the variables of our problem.

Example with conditional independence here

Bayes Nets

- We have a probabilistic model, which would require us to have the full joint table.
- We will describe the joint table using simple, local, conditional distributions
- We describe how the variables interact locally using a graph
 - part of a family called **probabilistic graphical models**
- Local interactions chain together to give global interactions

Example Bayes Net for diagnosing a car that won't start



Notice that there are relatively few dependencies between the nodes!

Notations

- Nodes: variables with domains
 - Can be assigned (observed) or unassigned (unobserved)
- Directed edges
 - Indicate **direct influence** between variables
 - Formally: encode conditional independence
- Although this is not strictly correct, you can imagine that the edges mean direct causation.

Example: coin flips

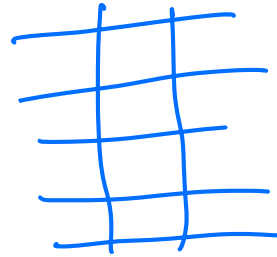
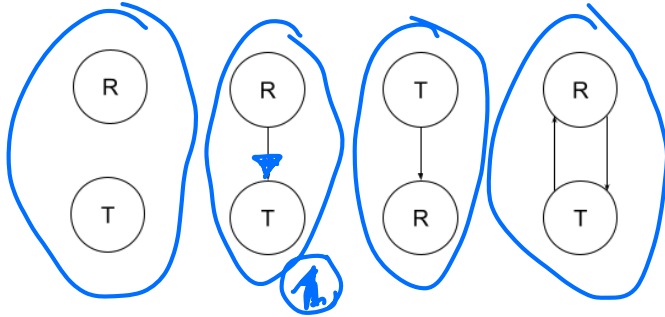
- N independent coin flips



- No interactions between variables: **absolute independence**

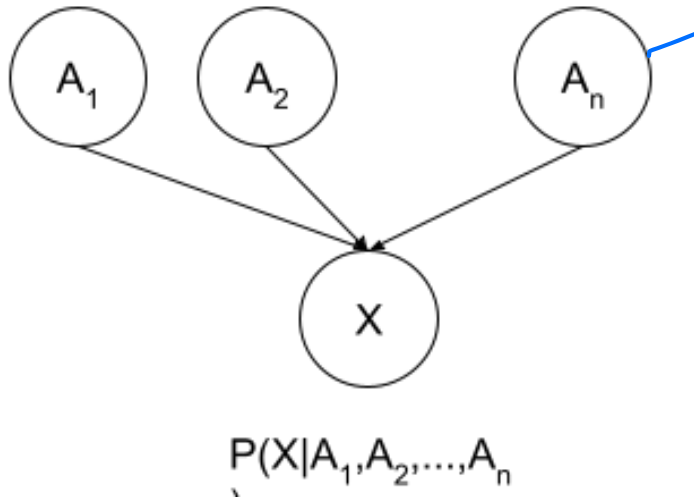
Example: traffic

- Variables: R it rains, T there is traffic
- We can model it as independent (model 0)
- Or we can model it that rain causes traffic (model 1) or traffic causes rain (model 2)



- Which is better? What does an agent gain by using model 1?

Bayes' net semantics



- A set of nodes, one per variable X
- A directed, acyclic graph
- A conditional distribution for each node
 - A collection of distributions over X , one for each combination of parent's values
 - $P(X|a_1, a_2, \dots, a_n)$
- **Bayes Net = Topology (graph) + Local Conditional Probabilities**

Probabilities in Bayes' nets

$\boxed{6,6}$ $\boxed{5,5}$
10%

- Bayes' nets implicitly encode joint distributions
 - As a product of local conditional distributions
 - To see what probability a Bayes Net gives to a full assignment, multiply all the relevant conditionals together


 $\frac{1}{36}$


 $\frac{1}{36}$

$\frac{1}{36 \times 36}$

$$P(x_1, x_2, \dots, x_n) = \prod_{i=1}^n P(x_i | \text{parents}(X_i))$$

- This means that a Bayes net cannot express any possible joint distribution, only those that can be written this way.
 - The smaller the number of arrows (parent/child relationships) the less number of values we need to describe the joint distro described by the Bayes net.

Probabilities in Bayes' nets

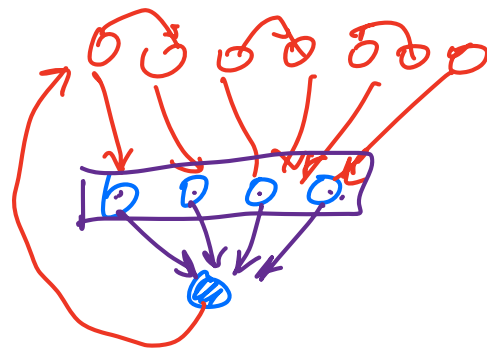
- But, wait, will this result in proper joint distribution? Meaning, does it add up to 1.0? ✓
- Chain rule (valid for all distributions)

$$P(x_1, x_2, \dots, x_n) = \prod_{i=1}^n P(x_i | x_1, \dots, x_{i-1})$$

- If we assume the conditional independence:

$$P(x_i | x_1, \dots, x_{i-1}) = P(x_i | \text{parents}(X_i))$$

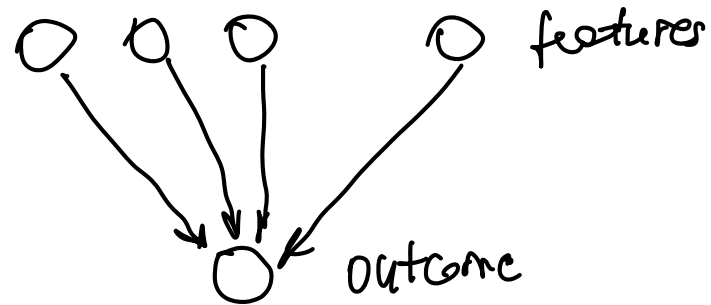
- What this means that the node only depends on the parents!
 - If we know how the parents turned out, the probability of the node doesn't depend on the rest of the network ("**d-separation**")



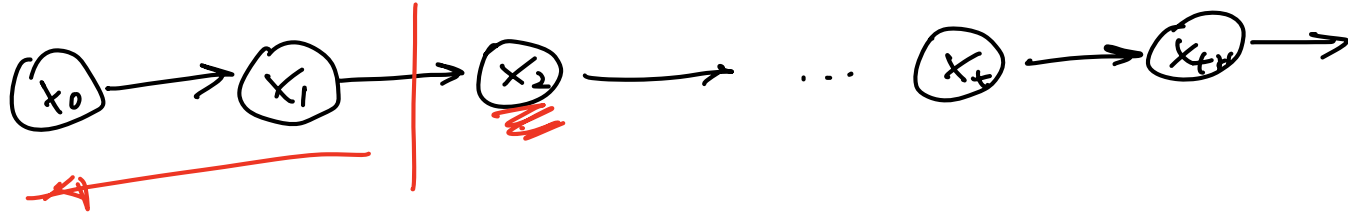
Example Bayes net - independent nodes



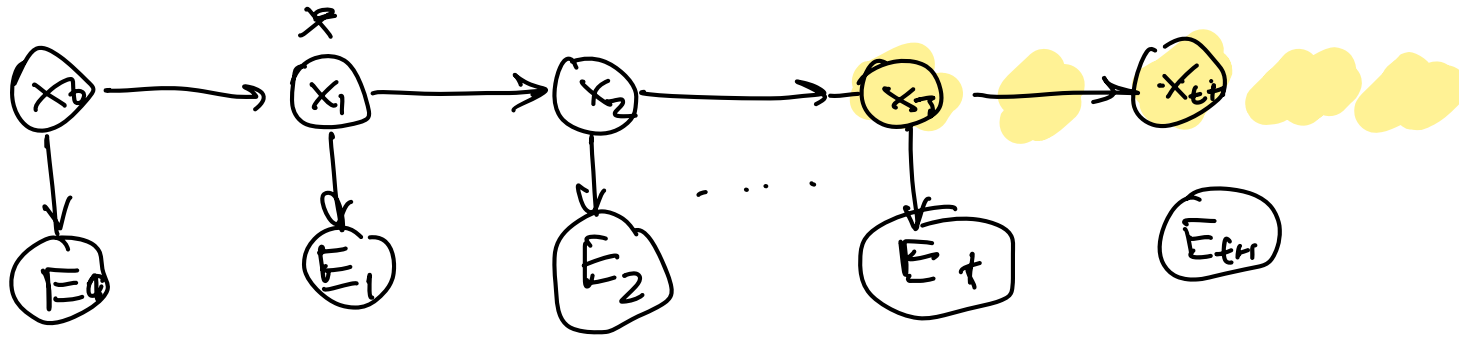
Example Bayes net - Naive Bayes



Example Bayes net - Markov chain



Example Bayes net - Hidden markov model



Causality?

- Bayes' nets are just conditional probabilities
 - Arrows just encode conditional independence
 - The arrows only reflect correlation, not causation.
 - This is good, because we can build a Bayes net even if we don't understand causation
- Sometimes we can build the Bayes net such that it really reflects causality
 - Often simpler (nodes have fewer patterns)
 - Often easier to think about
 - Often easier to elicit from experts

What do we think about Bayes nets?

- The whole area of probabilistic graphical models was the last great AI idea before the deep learning revolution
- It is still in the spirit of expert systems:
 - Interview an expert to deduct the structure of the Bayes net
 - Ask the expert for the probabilities - or learn the probabilities from data.
- They still have benefits:
 - Explainability: they can be understood by humans
 - Often learning can be more efficient if we first restrict what we are learning by recognizing conditional independencies.